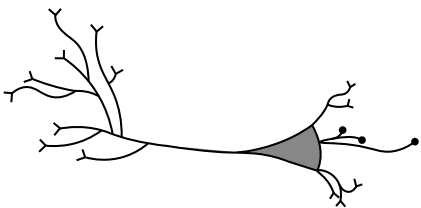
Recurrent dynamics and E–I balance

Wits Computational Neuroscience COMS4070A / COMS7081A: Lecture 5

August 19, 2026

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Where we are and where we are going

Lecture 4: What is the neural code and neural representation?

- Neural coding and tuning curves
- Spike-triggered average (STA)
- Generalised linear models (GLMs)
- Latent variable models (LVMs)

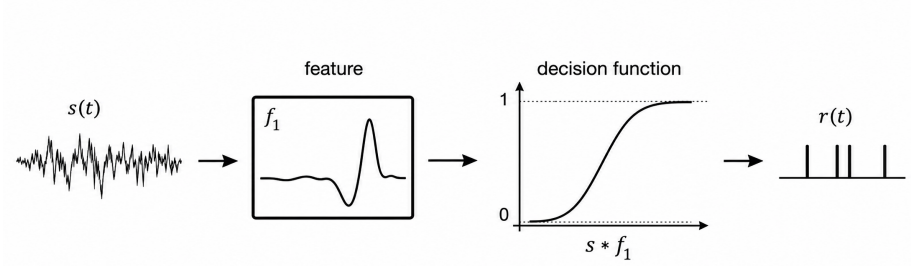
Lecture 5: How do neural representations evolve?

- Recurrent connectivity
- Excitatory–inhibitory interactions
- Dynamical systems

The network itself becomes part of the computation.

Population states are dynamical (1)

Last week: stimulus → (encoding) → population state



ChatGPT prompt, August 2026

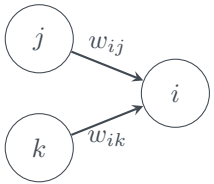
A stimulus can be represented by a **pattern of activity** $\boldsymbol{r}(s)$ across many neurons with tuning curves $f_i(s)$. *But this population state is not static!* E.g. Neural adaptation over time.

Population states are dynamical (2)

Last week we assumed that $\boldsymbol{r}(s)$ is the (population) representation of a stimulus. Where do temporal (e.g. neural adaptation) and other fluctuations come from?

Neurons are continuously receiving input from **other neurons**.

E.g. Why and how do seizures arise?



The activity at time t from other neurons contributes to determining the activity at time $t + \Delta t$.

$$\boldsymbol{r}(t) \quad \longrightarrow \quad \boldsymbol{r}(t + \Delta t)$$

How could we model this biologically?

For each of N neurons, we could explicitly model membrane potentials and synaptic conductances (inspired by lectures 2-3: LIF neurons and synaptic plasticity)...

$$\tau_m \frac{dV_i}{dt} = \underbrace{f_i(V_i, t)}_{\text{single neuron dynamics e.g. HH}} - \sum_{j \neq i} g_j(t) \underbrace{m_{ij}(V_i - E_j)}_{\text{synaptic weight, } := -w_{ij}}$$

$$\tau_s \frac{dg_j}{dt} = \underbrace{-g_j(t) + \sum_k \delta\left(t - t_k^{(j)}\right)}_{\text{synaptic conductance}} .$$

Describing

pre-synaptic spikes $\rightarrow$ synaptic input $\rightarrow V_i \rightarrow$ post-synaptic spikes

But this is hard to analyse

We now have a $2N$-dimensional dynamical system:

$$\underbrace{\{V_1, \ldots, V_N, g_1, \ldots, g_N\}}_{2N \text{ state variables}}$$

And it contains nonlinear interactions such as

$$\boxed{g_j(t)V_i(t)}$$

in addition to the nonlinear single-neuron dynamics $f_i(V_i, t)$.

Question: What kinds of collective dynamics can emerge?

Do we really need all this detail to study population dynamics?

We will simplify the model — while trying not to throw out the baby with the bathwater!

Zooming out: from neurons to population activity

The microscopic biological description has many details that we do not need.

Instead, **approximate** each neuron by its coarse-grained activity (temporal average):

$$V_i(t),\ \text{spikes} \quad \longrightarrow \quad r_i(t)$$

and collect the neurons into

$$\boldsymbol{r}(t) = \begin{bmatrix} r_1(t) \\ r_2(t) \\ \vdots \\ r_N(t) \end{bmatrix}$$

Then the recurrent connection (synaptic strength) weights become a matrix:

$$W = \begin{bmatrix} w_{11} & \cdots & w_{1N} \\ \vdots & \ddots & \vdots \\ w_{N1} & \cdots & w_{NN} \end{bmatrix}$$

A population-level recurrent dynamical system

At the population level, we can write

$$\tau \frac{d\boldsymbol{r}}{dt} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$$

where

- $\boldsymbol{r}$ = population activity
- W = recurrent connectivity
- $\boldsymbol{I}$ = external input
- ϕ = nonlinear input–output function



Today's question

What does the recurrent connectivity W do to the dynamics of the population?

And how does inhibition shape those dynamics?

The network itself becomes part of the computation

A purely feedforward view:



A recurrent view:



The response depends on

input + recurrent connectivity + network state

Key idea

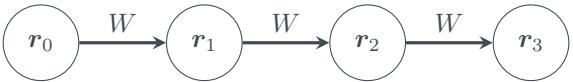
The network is not just **representing** information. Its dynamics can **transform and compute** that information.

Recurrence makes neural activity history-dependent

Because

$$\boldsymbol{r}(t) \to W\boldsymbol{r}(t) \to \boldsymbol{r}(t + \Delta t),$$

the current state depends on previous states.



The network's future depends on its current state.

A simple toy model

$$\boxed{\tau \frac{d\boldsymbol{r}}{dt} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})}$$

To get some intuition, let us define

$$\phi(x) = x.$$

Then assuming we just have a single neuron,

$$\tau \frac{dr}{dt} = -r + wr + I = (w-1)r + I.$$

Exercise: What is the stable state?

At a fixed point, $\frac{dr}{dt} = 0$.

Therefore:

$$0 = (w-1)r^* + I$$

$$\iff (1-w)r^* = I$$

and hence

$$\boxed{r^* = \frac{I}{1-w}}$$

Question

What happens as recurrent excitation w increases?

What does **increasing** recurrent excitation do?

Consider the input:

The fixed point (F.P.) is $r^* = \frac{I}{1-w}$.
For $0 \leq w < 1$:

w	r^*/I
0	1
0.5	2
0.9	10

The network can **amplify** an external input.

Consider the timescale:
Consider a small perturbation around the F.P.:

$$r(t) = r^* + \delta r(t).$$

Then $$\tau \frac{d(\delta r)}{dt} = (w-1)\delta r, \qquad \text{and}$$

$$\delta r(t) = \delta r(0) \exp\left[-\frac{1-w}{\tau} t\right].$$

So the effective time constant (= solving for τ_{eff} in the usual exponential decay equation $\delta r(t) = \delta r(0) e^{-t/\tau_{\text{eff}}}$) is

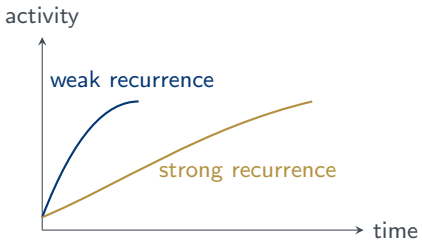
$$\boxed{\tau_{\text{eff}} = \frac{\tau}{1-w}} \qquad (w < 1).$$

The network can **increase** the time constant.

Recurrence changes gain and timescale

As w increases toward 1:

gain $\uparrow$ $\quad$ τ_{eff} $\uparrow$



Recurrent excitation can make a network both **more sensitive** and **slower**.

Are there any other possible regimes or *problems* with this model (e.g. consider other parameter values)?

What happens with $w > 1$?

Recall

$$\tau \frac{d(\delta r)}{dt} = (w-1)\delta r.$$

If $w > 1$, then

$$\delta r(t) \propto e^{+(w-1)t/\tau}.$$

Recurrent excitation can become **unstable**.

Why does inhibition matter?

Excitation $\longrightarrow$ amplification

Too much excitation $\longrightarrow$ instability

Excitation + inhibition $\longrightarrow$ controlled dynamics

Question: How do excitatory and inhibitory populations interact to shape the dynamics?

Excitation and inhibition: A two-population E–I network

Excitation (depolarisation)

- increases downstream activity
- primarily glutamatergic

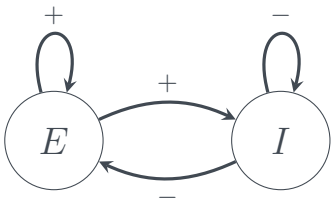
Inhibition (hyperpolarisation)

- decreases downstream activity
- primarily GABAergic

$$I_{\text{syn}} = I_{\text{E}} + I_{\text{I}}$$

Let $r_{\text{E}}(t)$ and $r_{\text{I}}(t)$ denote excitatory and inhibitory population activity.

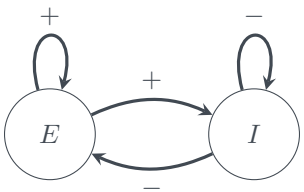
Each population contributes to the other's future activity.



The E–I rate model

$$\tau_{\mathrm{E}} \dot{r}_{\mathrm{E}} = -r_{\mathrm{E}} + \phi\left(w_{\mathrm{EE}} r_{\mathrm{E}} - w_{\mathrm{EI}} r_{\mathrm{I}} + I_{\mathrm{E}}\right)$$

$$\tau_{\mathrm{I}} \dot{r}_{\mathrm{I}} = -r_{\mathrm{I}} + \phi\left(w_{\mathrm{IE}} r_{\mathrm{E}} - w_{\mathrm{II}} r_{\mathrm{I}} + I_{\mathrm{I}}\right)$$



Fixed points of the E–I system

For intuition, let's assume a linear input–output function $\phi(x) = x$ with equal time constants ($\tau = 1$):
Then

$$\dot{r}_{\mathrm{E}} = -r_{\mathrm{E}} + w_{\mathrm{EE}} r_{\mathrm{E}} - w_{\mathrm{EI}} r_{\mathrm{I}} + I_{\mathrm{E}},$$

$$\dot{r}_{\mathrm{I}} = -r_{\mathrm{I}} + w_{\mathrm{IE}} r_{\mathrm{E}} - w_{\mathrm{II}} r_{\mathrm{I}} + I_{\mathrm{I}}.$$

Exercise: How do we find the fixed point?

At the F.P. $\dot{r}_{\mathrm{E}} = \dot{r}_{\mathrm{I}} = 0$.
So the network satisfies

$$0 = (w_{\mathrm{EE}} - 1) r_{\mathrm{E}} - w_{\mathrm{EI}} r_{\mathrm{I}} + I_{\mathrm{E}}$$

and

$$0 = w_{\mathrm{IE}} r_{\mathrm{E}} - (1 + w_{\mathrm{II}}) r_{\mathrm{I}} + I_{\mathrm{I}}.$$

Question

What are these equations called? Where in $(r_{\mathrm{E}}, r_{\mathrm{I}})$ space can both equations be satisfied?

An example E–I network

Let $w_{\mathrm{EE}} = 0.5, \quad w_{\mathrm{EI}} = 1,$

$w_{\mathrm{IE}} = 1, \quad w_{\mathrm{II}} = 0.5,$

and

$I_{\mathrm{E}} = 1, \quad I_{\mathrm{I}} = 0.5.$

Exercise: What is the value of the F.P.?

$\dot{r}_{\mathrm{E}} = -0.5r_{\mathrm{E}} - r_{\mathrm{I}} + 1$ and

$\dot{r}_{\mathrm{I}} = r_{\mathrm{E}} - 1.5r_{\mathrm{I}} + 0.5.$

At the fixed point:

$r_{\mathrm{I}} = 1 - \frac{1}{2}r_{\mathrm{E}}, \quad$ and $\quad r_{\mathrm{I}} = \frac{2}{3}r_{\mathrm{E}} + \frac{1}{3}.$

These are the two nullclines.

The fixed point is where the dynamics agree

For this example,

$$\dot{r}_\mathrm{E} = 0 \quad \Longrightarrow \quad r_\mathrm{I} = 1 - \frac{1}{2} r_\mathrm{E},$$

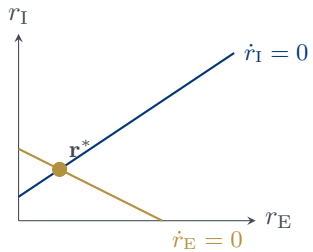
$$\dot{r}_\mathrm{I} = 0 \quad \Longrightarrow \quad r_\mathrm{I} = \frac{2}{3} r_\mathrm{E} + \frac{1}{3}.$$

Setting them equal:

$$1 - \frac{1}{2} r_\mathrm{E} = \frac{2}{3} r_\mathrm{E} + \frac{1}{3}$$

so

$$r_\mathrm{E}^* = \frac{4}{7}, \qquad r_\mathrm{I}^* = \frac{5}{7}.$$



fixed point = intersection of the nullclines

The population is a dynamical system

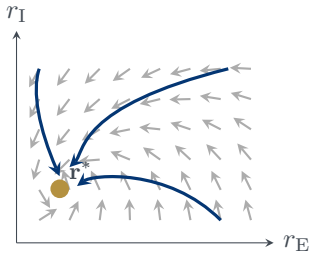
The network is a point in state space:

$$\boldsymbol{r}(t) = \begin{pmatrix} r_{\mathrm{E}}(t) \\ r_{\mathrm{I}}(t) \end{pmatrix}.$$

Over time,

$$\boldsymbol{r}(t_0) \to \boldsymbol{r}(t_1) \to \boldsymbol{r}(t_2) \to \cdots$$

The equations tell us not only where the F.P. is, but also which way the state moves.



A recurrent network generates a **trajectory through state space**. In this example, $\mathbf{r}^\star$ is a **stable fixed point**. How do you determine the stability of F.P.s?

What does E–I interaction actually accomplish?

Stabilise

$E \uparrow \rightarrow I \uparrow \rightarrow E \downarrow$

Amplify

$E \uparrow \rightarrow E \uparrow$

when recurrent excitation dominates locally

Compete

$E_1 \uparrow \rightarrow I \uparrow \rightarrow E_2 \downarrow$

The same external input can produce different responses depending on the network state (e.g. differing gain, input-output functions).

Large networks are often **sparse**

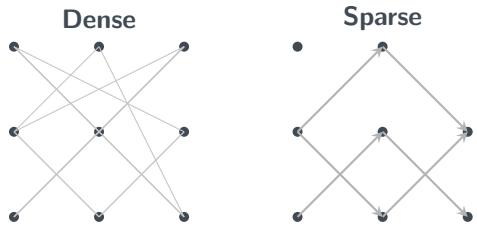
Suppose we have $N \gg 1$ neurons.

A neuron does *not* necessarily connect to every other neuron.

$$\boxed{K \ll N}$$

where K is the number of recurrent inputs per neuron.

Large population $\neq$ **dense connectivity**

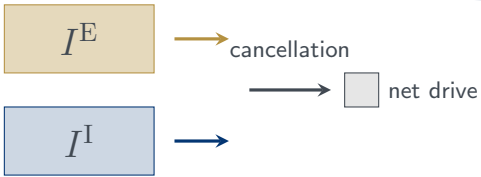


Here: large, *sparsely connected* recurrent networks.

Large excitation and inhibition can **cancel**

A neuron receives many excitatory and inhibitory inputs:

$$I_i = \underbrace{\sum_{j\in \mathrm{E}} w_{ij}^{\mathrm{E}} r_j^{\mathrm{E}}}_{I_i^{\mathrm{E}}} + \underbrace{\sum_{j\in \mathrm{I}} w_{ij}^{\mathrm{I}} r_j^{\mathrm{I}}}_{I_i^{\mathrm{I}}} + I_i^{\mathrm{ext}}.$$



In a balanced large-network regime,

$$I_i^{\mathrm{E}} \sim +\sqrt{K}, \qquad I_i^{\mathrm{I}} \sim -\sqrt{K}.$$

$$\boxed{I_i^{\mathrm{E}} + I_i^{\mathrm{I}} \sim O(1)}$$

Balanced network intuition

Large opposing synaptic inputs can nearly cancel, leaving a comparatively small net input.

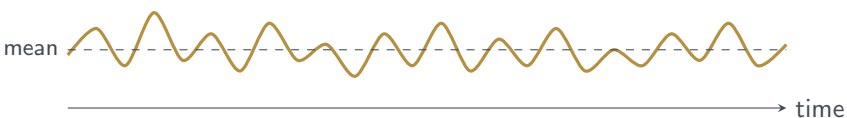
van Vreeswijk & Sompolinsky, *Science*, 1996

Balanced does **not** mean silent

$$\underbrace{\text{large excitation}}_{\text{ongoing}} + \underbrace{\text{large inhibition}}_{\text{ongoing}} \longrightarrow \underbrace{\text{small net input}}_{\text{on average}}$$

But the inputs are still fluctuating:

$$\boxed{\text{large E} + \text{large I} \longrightarrow \text{small mean} + \text{ongoing fluctuations}}$$



The network can remain active and irregular even when E and I largely cancel.

So far, we have used a linear dynamical system

We assumed $\phi(x) = x$, so the dynamics are essentially

$$\boxed{\dot{\mathbf{r}} = A\mathbf{r} + \mathbf{I}}$$

for some matrix A.

This made the system easy to analyse:

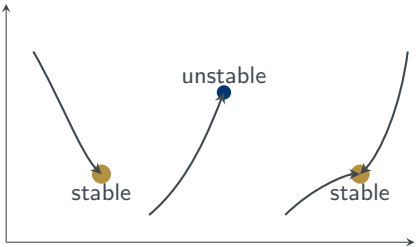
nullclines $\rightarrow$ F.P. $\rightarrow$ vector field $\rightarrow$ stability

Neural input–output functions are non-linear.

What changes when we restore ϕ?

$$\tau \frac{d\boldsymbol{r}}{dt} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$$

Non-linear ϕ can "bend" dynamics in state space.



What if there is more than one stable state?

A nonlinear recurrent network can have

multiple stable fixed points.

Then different initial states can converge to different stable states.

initial state $\longrightarrow$ stable state 1

or

initial state $\longrightarrow$ stable state 2

Next lecture

What can multiple stable states let a neural network *compute*?

Attractors, decisions and memory.

How can we analyse a nonlinear system?

Suppose

$$\boxed{\dot{\boldsymbol{r}} = \phi(\boldsymbol{r})}$$

and the network sits at a fixed point

$$\phi(\boldsymbol{r}^*) = 0.$$

Consider a small perturbation:

$$\boldsymbol{r}(t) = \boldsymbol{r}^* + \delta\boldsymbol{r}(t).$$

Taylor expansion gives

$$\phi(\boldsymbol{r}^* + \delta\boldsymbol{r}) \approx \phi(\boldsymbol{r}^*) + \left.\frac{\partial\phi}{\partial\boldsymbol{r}}\right|_{\boldsymbol{r}^*} \delta\boldsymbol{r}.$$

Since $\phi(\boldsymbol{r}^*) = 0$,

$$\boxed{\frac{d\,\delta\boldsymbol{r}}{dt} \approx J\,\delta\boldsymbol{r}}$$

where

$$\boxed{J = \left.\frac{\partial\phi}{\partial\boldsymbol{r}}\right|_{\boldsymbol{r}^*}}$$

is the **Jacobian**.

The Jacobian of the E–I network (1)

For the Wilson–Cowan system,

$$G_E := \dot{r}_\mathrm{E} = \frac{1}{\tau_E}\left[-r_\mathrm{E} + \underbrace{\phi(w_{\mathrm{EE}} r_\mathrm{E} - w_{\mathrm{EI}} r_\mathrm{I} + I_\mathrm{E})}_{:=\phi_E}\right],$$

$$G_I := \dot{r}_\mathrm{I} = \frac{1}{\tau_\mathrm{I}}\left[-r_\mathrm{I} + \underbrace{\phi(w_{\mathrm{IE}} r_\mathrm{E} - w_{\mathrm{II}} r_\mathrm{I} + I_\mathrm{I})}_{:=\phi_I}\right].$$

$$J = \begin{pmatrix} \frac{\partial}{\partial r_E} G_E(r_E^\star, r_I^\star) & \frac{\partial}{\partial r_I} G_E(r_E^\star, r_I^\star) \\ \frac{\partial}{\partial r_E} G_I(r_E^\star, r_I^\star) & \frac{\partial}{\partial r_I} G_I(r_E^\star, r_I^\star) \end{pmatrix}$$

Linearising around a fixed point gives

$$\frac{d}{dt}\begin{bmatrix}\delta r_\mathrm{E} \\ \delta r_\mathrm{I}\end{bmatrix} = J \begin{bmatrix}\delta r_\mathrm{E} \\ \delta r_\mathrm{I}\end{bmatrix}.$$

e.g. $$\frac{\partial}{\partial r_E} G_E(r_E^\star, r_I^\star) = \frac{1}{\tau_E}\big[-1 + \underbrace{w_{EE}\phi_E'(r_E^\star, r_I^\star)}_{\text{chain rule}}\big]$$

$$= \frac{1}{\tau_E}\left[-1 + w_{EE}\phi'(w_{\mathrm{EE}} r_\mathrm{E}^\star - w_{\mathrm{EI}} r_\mathrm{I}^\star + I_\mathrm{E})\right]$$

The Jacobian of the E–I network (2)

For the Wilson–Cowan system,

$$\tau_{\mathrm{E}}\dot{r}_{\mathrm{E}} = -r_{\mathrm{E}} + \phi(w_{\mathrm{EE}}r_{\mathrm{E}} - w_{\mathrm{EI}}r_{\mathrm{I}} + I_{\mathrm{E}}),$$
$$\tau_{\mathrm{I}}\dot{r}_{\mathrm{I}} = -r_{\mathrm{I}} + \phi(w_{\mathrm{IE}}r_{\mathrm{E}} - w_{\mathrm{II}}r_{\mathrm{I}} + I_{\mathrm{I}}).$$

Linearising around a fixed point gives

$$\frac{d}{dt}\begin{bmatrix}\delta r_{\mathrm{E}} \\ \delta r_{\mathrm{I}}\end{bmatrix} = J\begin{bmatrix}\delta r_{\mathrm{E}} \\ \delta r_{\mathrm{I}}\end{bmatrix}.$$

$$J = \begin{pmatrix} \dfrac{-1 + w_{\mathrm{EE}}\phi'_{\mathrm{E}}}{\tau_{\mathrm{E}}} & \dfrac{-w_{\mathrm{EI}}\phi'_{\mathrm{E}}}{\tau_{\mathrm{E}}} \\ \dfrac{w_{\mathrm{IE}}\phi'_{\mathrm{I}}}{\tau_{\mathrm{I}}} & \dfrac{-1 - w_{\mathrm{II}}\phi'_{\mathrm{I}}}{\tau_{\mathrm{I}}} \end{pmatrix}_{\boldsymbol{r}=\boldsymbol{r}^*}$$

Connectivity + gain determine local dynamics.

The eigenvalues tell us what happens

For

$$\frac{d\,\delta \boldsymbol{r}}{dt} = J\,\delta \boldsymbol{r},$$

suppose

$$J\boldsymbol{v} = \lambda \boldsymbol{v}.$$

Then a perturbation along $\boldsymbol{v}$ evolves as

$$\delta \boldsymbol{r}(t) \propto e^{\lambda t}\boldsymbol{v}.$$

Therefore:

$$\boxed{\mathrm{Re}(\lambda) < 0 \quad \Rightarrow \quad \text{perturbations decay}}$$

$$\boxed{\mathrm{Re}(\lambda) > 0 \quad \Rightarrow \quad \text{perturbations grow}}$$

Complex eigenvalues produce **oscillations**

Suppose the Jacobian has eigenvalues and

$$\boxed{\lambda = \alpha \pm i\omega}$$

$\omega \quad \Rightarrow \quad$ oscillation frequency

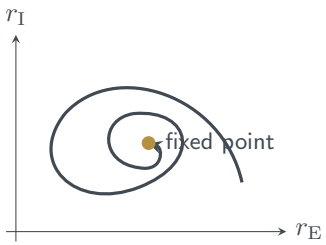
Then perturbations behave like

$$\delta \boldsymbol{r}(t) \sim e^{\alpha t} \begin{pmatrix} \cos(\omega t) \\ \sin(\omega t) \end{pmatrix}.$$

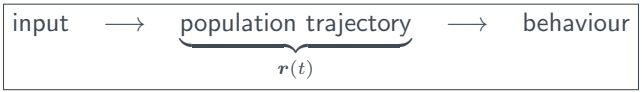
So:

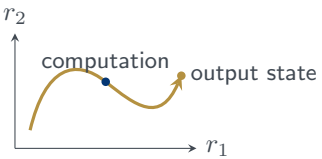
$\alpha < 0 \quad \Rightarrow \quad$ damped oscillations

$\alpha > 0 \quad \Rightarrow \quad$ growing oscillations



Applied example: Dynamics can implement a computation (1)

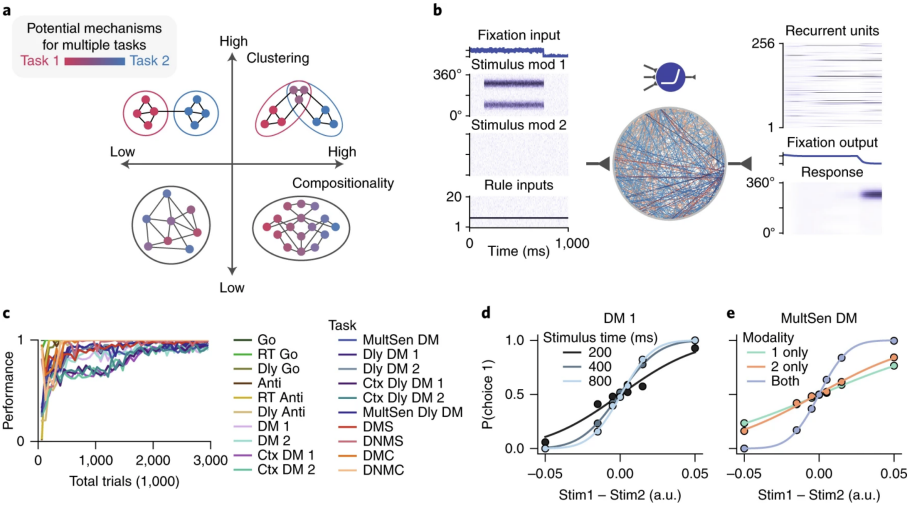




The computation is not just the instantaneous neural state. **The trajectory through state space can carry out the computation.**

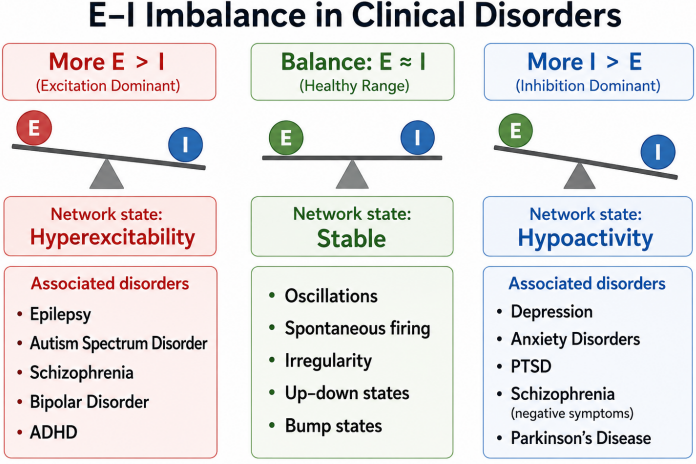
Yang et al., *Nat Neurosci*, 2019

Applied example: Dynamics can implement a computation (2)



Yang et al., *Nat Neurosci*, 2019

Applications of E-I balance in psychiatry



ChatGPT prompt, August 2026

Wertheimer and Hart, *Nat Neurosci*, 2024; Jardri and Denéve, *Brain*, 2013; Gao et al., *NeuroImage*, 2017

If we have time...

Another way to analyse large networks

So far, we have tracked the trajectory of the population state $\boldsymbol{r}(t)$.

What if we care instead about **population statistics**?

Extension 1: From trajectories to statistics

So far we have tried to understand $\boldsymbol{r}(t)$, the activity of every neuron at every moment.

But with many neurons, perhaps we should ask statistical questions.

For each neuron:

$$r_i(t) = \underbrace{\nu_i}_{\text{mean activity}} + \underbrace{\delta r_i(t)}_{\text{fluctuations}}$$

This suggests two different questions:

1. How does activity vary **across neurons**?
2. How does activity vary **over time**?

Instead of solving every neural trajectory — high-dimensional dynamics — we might ask about

$$P(\nu)$$

the distribution of firing rates across neurons.

individual trajectories	$\longrightarrow$	population statistics

This is the basic idea behind mean-field approaches.

Extension 2: What about the fluctuations?

We separated activity into

$$r_i(t) = \nu_i + \delta r_i(t).$$

The **mean** tells us about the population's average state, but the fluctuations contain temporal structure.

$$\delta r_i(t) \quad \longrightarrow \quad \text{temporal correlations}$$

For example:

$$C(\Delta t) = \langle \delta r_i(t) \delta r_i(t + \Delta t) \rangle \, .$$

Fluctuations

How does recurrent connectivity shape the timescale and structure of population fluctuations?

Take home and resources

Question	Idea	Consequence
What is a population state?	$\boldsymbol{r}(t)$	A trajectory in state space
What does recurrence do?	Feedback	Gain + timescales
Why is inhibition important?	E–I interactions	Stability + competition
What happens in large networks?	Sparse + balanced inputs	Ongoing fluctuations
What does nonlinearity add?	Local stability	Multiple states + oscillations
Why do dynamics matter?	Population trajectories	Computation

Resources

- Neuromatch Wilson-Cowan tutorial
- Neuromatch Wilson-Cowan extension

- Denéve and Machens, 2016, "Efficient codes and balanced networks (Review)"
- Yang et al., 2019, "Task reps in neural nets trained to perform many cognitive tasks"

Next up: attractors, decisions and memory

Recurrent dynamics

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Stable population states

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Multiple stable states

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Attractors